

Yiğit Sarp Avcı

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Education

Boğaziçi University - B.Sc. in Computer Engineering

09/2023 – Present

Kabataş Boys' High School

09/2018 – 06/2023

- Graduated with a GPA of 3.78/4.00.
- Actively participated in various extracurricular activities and student societies, including Debate, Model United Nations, Entrepreneurship, and the Coding Club.

Experience

Executive Board Member at Radyo Boğaziçi

09/2023 – Present

- Developed strong skills in team coordination, social media management, budget planning, event organization, communication, and operational decision-making.

Work and Travel Program Participant

06/2025 – 09/2025

- Gained valuable international experience through the cultural exchange and work program in the USA which improved my communication, English, teamwork and problem-solving skills.

Regional Organizational Team Member at ESTIEM

09/2023 – Present

- Contributed to the planning and coordination of events, team communication, and organizational processes within ESTIEM, an international network connecting Industrial Engineering and Management students across Europe.

Technical Projects

AI-Powered FinTech Intelligence Platform — Next.js, FastAPI, PostgreSQL, Docker

- Architected a full-stack, deployment-ready financial dashboard combining personal finance analytics, an algorithmic portfolio risk evaluator, and OpenAI-generated insights with secure NLP fallbacks.
- Engineered a live limit-order trading simulator featuring a synchronous Price-Time Priority matching engine and achieved 100% backend test coverage with automated CI/CD pipelines.

SchedSim CPU Scheduler — x86-64 Assembly, GNU Toolchain, Linux Syscalls

- High-performance CPU scheduling simulation engine (FCFS, SJF, SRTF, PF, RR) built with zero standard library dependencies, utilizing direct Linux syscalls for I/O and process management.

Spirelayer Command Interpreter — C, Custom Backtracking Lexer

- Robust CLI-based interpreter replicating fantasy RPG REPL mechanics; implemented an autonomous backtracking lexical parser to handle non-deterministic grammar and complex command resolution.

Aura Functional Interpreter — Python, Functional Paradigms, Dual Scoping Engine

- Advanced interpreter for a custom functional language supporting first-class closures, higher-order function patterns, and toggleable lexical/dynamic scoping engines for complex variable resolution.

Technical Skills

- **Languages:** C, C++, Java, Python, TypeScript, JavaScript, x86-64 Assembly, VHDL, Bash
- **Web & Cloud Architecture:** FastAPI, Next.js, React, PostgreSQL, Docker, REST APIs, CI/CD
- **Low-Level & Systems:** Pthreads, Linux Syscalls, Memory Management, Process Scheduling, FPGA
- **AI & Modern Engineering:** Vibecoding, Autonomous Coding, LLM Prompting, AI-Assisted Architecture
- **Core Theory:** Data Structures & Algorithms, Complexity Analysis (Big O), Graph Theory, Discrete Math
- **Software & Ecosystem:** Linux/Unix, Git, GitHub, LaTeX, MS Excel, PowerPoint, Canva, Adobe Photoshop

Languages

- Turkish (native)
- English (fluent)
- German (elementary)

Extracurricular Activities

- DJ, Radyo Boğaziçi
- Student-Athlete, Boğaziçi University Sailing Team
- Student-Athlete, Boğaziçi University Ski Team
- Windsurf Athlete, Competing Individually